

# Avnish Dubey

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(Open to Relocate)

## PROFESSIONAL SUMMARY

Experienced **Java Full-Stack** with **5+** years of expertise in building secure, scalable payment systems leveraging robust back-end frameworks (**Spring Boot, Spring JPA, Spring Security**) and **API integrations**. Skilled in designing **RESTful** and **GraphQL** APIs, architecting microservices, and orchestrating data workflows using **Apache Airflow**. Strong experience with **SQL** and **NoSQL** databases, along with implementing DevOps practices using **Docker, CI/CD** and testing frameworks to ensure reliable delivery.

## WORK EXPERIENCE

### Lam Research Java Full-Stack

**Fremont, CA**  
**Apr 2025 – Present**

*Migrated Solr-based APIs to PostgreSQL and built GraphQL APIs with Spring Boot to modernize backend data access, cutting latency and improving concurrency.*

- Migrated Solr-based **REST APIs** to **PostgreSQL** using native queries, reducing latency by **35%**, and improving data consistency by **80%**, while aligning the service with enterprise cloud modernization goals.
- Developed scalable backend applications using **Java and Spring Boot**, building **25+ REST and GraphQL** APIs, optimizing database interactions, and delivering high-performance enterprise services supporting **100K+ API requests** per day.
- Built modular backend components following **Spring MVC** architecture, enabling secure, high-performance request handling and database integration.
- Developed Java Spring Boot backend services deployed on **AWS**, contributing to scalable, secure, and highly available enterprise applications with **99.9% service availability**.
- Containerized **20+ Java Spring Boot and GraphQL** applications using **Docker**, ensuring consistent development, testing, and deployment across environments.
- Deployed and managed containerized backend services on **Kubernetes**, enabling scalable, highly available, and resilient application deployments supporting **50+ production deployments**.
- Utilized **Postman** to develop, test, and validate **REST** and **GraphQL** APIs, ensuring functional accuracy and reliable backend integrations.
- Devised comprehensive **JUnit** test cases to validate multi-value filtering logic and PostgreSQL query accuracy, improving test coverage by **25%** and preventing critical data mismatches across **backend APIs**.
- Designed and implemented **GraphQL APIs** with **Spring Boot** to enable flexible, client-driven data retrieval for near real-time workflows, reducing redundant backend requests and improving **API response** efficiency by **30–40%**.
- Implemented high-performance GraphQL resolvers integrated with **SQL** databases and backend **microservices**, improving **query execution** efficiency by **30%** and supporting **1K+ concurrent** client requests consistent response **SLAs**.

### Principal Financial Software Engineer I

**New York, NY**  
**June 2024 - Mar 2025**

*Developed RESTful and GraphQL APIs for Stripe payment processing, integrated Kafka event-driven systems, and automated cloud workflows using Airflow and GCP*

- Developed dedicated **RESTful** and **GraphQL APIs** for payment request validation and routing to **Stripe**, accomplishing a **25%** improvement in processing efficiency
- Integrated GraphQL services with event-driven backend systems (**Kafka** and **microservices**), enabling low-latency data access and improving downstream data processing efficiency by **20–25%**
- Automated WDL file parsing and ingestion workflows by orchestrating containerized jobs on Google Cloud Run through **Apache Airflow**, leveraging **GCS** and Cloud Build to reduce manual intervention by **70%** and significantly improve data reliability.
- Designed and delivered GraphQL APIs on top of **Spring Boot** payment services, enabling flexible client-driven data retrieval and improving payment processing efficiency by **20–25%** by reducing redundant service calls

### Tata Consultancy Services System Engineer (Java)

**Pune, India**  
**Dec 2021 – Jul 2022**

*Developed Spring MVC/Boot APIs for real-time market data aggregation and historical performance simulation, addressing risk analysis and processing speed needs for a financial platform.*

- Designed dual Spring **MVC APIs**—one aggregating live market data via third-party APIs with **Redis** caching for risk analysis and another simulating historical performance using asynchronous processing—boosting processing speed by **30%**.
- Optimized **PostgreSQL** database performance in a Spring Boot application by leveraging **Hibernate ORM**, query optimization, and **Redis** caching, while integrating performance monitoring dashboards, improving API response times by **15%**.
- Optimized application delivery by containerizing services with **Docker**, orchestrating them on **Kubernetes**, and implementing automated **CI/CD pipelines**, which cut release cycles by **40%**.
- Utilized **socket** programming techniques to boost system scalability, resulting in an **80%** enhancement

## TopHat Software Technologies Software Engineer

**Indore, India**  
**Feb 2020 – Nov 2021**

*Built Spring Boot/Kafka alerts and tax APIs, created JUnit/TestNG/JMeter testing suite, and automated workflows with UiPath RPA.*

- Spearheaded development of a **Java Spring Boot**–based Alerts API using **Kafka** for real-time notifications and a Tax Insights API for dynamic tax analysis, enabling investors to react **25%** faster to market changes.
- Formulated a robust testing suite combining **unit tests (JUnit)**, integration tests (TestNG), and performance tests (JMeter) for the manufacturing system, leading to a **40%** drop in critical bugs
- Automated robots/workflows using UI Path (Robotic Process Automation), scraping data from websites, resulting in a **60% reduction** in manual data entry efforts and saving **100% of manual effort time**.

## PROJECTS

### OpsPilot — AI-Powered Incident Management Platform

*Java 21, Spring Boot, PostgreSQL, Spring Security, Kafka, Spring AI, Ollama, Docker*

- Built a secure incident management platform using Java, Spring Boot, PostgreSQL, and Spring Security
- Developed an event-driven alert pipeline processing **1,000+** simulated alerts per minute with idempotent Kafka consumers.
- Integrated **SpringAI** and **Ollama** to generate root-cause summaries and troubleshooting recommendations in under **10 seconds**.
- Implemented **JWT** authentication, RBAC, Flyway, **Docker**, and Testcontainers, achieving **80%+** test coverage and reducing investigation time by **40%**.

### PayGuard AI — Intelligent Payment & Risk Microservices Platform

*Java 21, Spring Boot, GraphQL, PostgreSQL, Docker, Spring AI, Kafka*

- Built an **AI-integrated payment** processing platform using **Java 21, Spring Boot, GraphQL, PostgreSQL**, and Docker, processing **1,000+** simulated transactions per minute.
- Designed **Kafka**–based fraud–scoring and audit pipelines, reducing synchronous processing overhead by **35%** while enabling near–real–time anomaly detection.
- Integrated **Spring AI/LangChain4j** to generate human–readable fraud explanations, reducing simulated transaction–review time by **45%**.
- Developed a natural–language payment–status assistant using **GraphQL** and **LLM**–powered queries, improving support–query response time by **60%**.

## SKILLS

- **Programming Languages:** Java (21,17, 11, 8), SQL, Node.js, Python, C++, Ruby, Go, JavaScript
- **Web Technologies:** HTML5, CSS3, AJAX, JSON, XML, Tailwind CSS, RESTful API, GraphQL
- **Cloud:** Google Cloud Platform, AWS (EC2, S3, DynamoDB, Lambda), Microsoft Azure
- **Frameworks:** React.js, Angular.js, Spring Boot, Bootstrap, Spring MVC, jQuery, Django, JSP, JPA, JavaBeans, JDBC, JVM
- **Database & Data Management:** MS SQL Server, Oracle, MySQL, MongoDB, PostgreSQL, NoSQL, DynamoDB, Firebase
- **DevOps & Version control:** Jenkins, Bitbucket, GitHub, GitLab, Docker, Kubernetes, Terraform
- **Development Tools:** Selenium, JUnit, Jest, Mockito, Swagger UI, Microservices, Jira, Confluence, IntelliJ IDEA, UiPath, Postman

EDUCATION

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|--------------------------------|---------------------|
| <b>Syracuse University</b>     | <b>Syracuse, NY</b> |
| Master's in Computer Science   | Aug 2022 - May 2024 |
| <b>Medi-Caps University</b>    | <b>Indore, IN</b>   |
| Bachelor's in Computer Science | Aug 2016 - May 2020 |